Estimations Combined

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Description

This code loads and combines the different feature value estimations (natural language writing, paleolithic signs, and proto-cuneiform) into one csv file.

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: install.packages("ggplot2").

```
#library(stringr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

Load Data

Load csv files with quantitative feature values for sign sequences.

SIGNBASE

```
# load signbase estimations for strings on objects
sign.data <- read.csv("~/Github/UpperPalCombinatorics/results/features/signBase_features.csv")
# select Aurignacian objects
sign.data <- sign.data[sign.data$archaeological_unit == "Aurignacien", ]

# select columns for direct comparison with TeDDi sample
sign.data.selected <- select(sign.data, c('archaeological_unit', 'coding_clean',
                                          'num.char', 'huni.chars', 'hrate.chars',
                                 'ttr.chars', 'rm.chars'))
# change column names
colnames(sign.data.selected) <- c('subcorpus', 'sequence', 'num.char','huni.chars', 'hrate.chars',
                          'ttr.chars', 'rm.chars')
```

CDLI

```r
# read feature values for proto-cuneiform from CDLI
# Uruk V
cuneiV.data <- read.csv("~/Github/UpperPalCombinatorics/results/features/cuneiform_UrukV_features.csv")

# add column with subcorpus information
cuneiV.data$subcorpus <- rep("Cuneiform (Uruk V)", nrow(cuneiV.data))

# select columns for direct comparison with other samples
cuneiV.data.selected <- select(cuneiV.data, c('subcorpus', 'signs.expand.vec',
                                              'num.char', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(cuneiV.data.selected)[names(cuneiV.data.selected) == "signs.expand.vec"] <- "sequence"

# Uruk IV
cuneiIV.data <- read.csv("~/Github/UpperPalCombinatorics/results/features/cuneiform_UrukIV_features.csv

# add column with subcorpus information
cuneiIV.data$subcorpus <- rep("Cuneiform (Uruk IV)", nrow(cuneiIV.data))

# select columns for direct comparison with other sample
cuneiIV.data.selected <- select(cuneiIV.data, c('subcorpus', 'signs.expand.vec',
                                              'num.char', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(cuneiIV.data.selected)[names(cuneiIV.data.selected) == "signs.expand.vec"] <- "sequence"

# Uruk III
cuneiIII.data <- read.csv("~/Github/UpperPalCombinatorics/results/features/cuneiform_UrukIII_features.c

# add column with subcorpus information
cuneiIII.data$subcorpus <- rep("Cuneiform (Uruk III)", nrow(cuneiIII.data))

# select columns for direct comparison with other sample
cuneiIII.data.selected <- select(cuneiIII.data, c('subcorpus', 'signs.expand.vec',
                                              'num.char', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(cuneiIII.data.selected)[names(cuneiIII.data.selected) == "signs.expand.vec"] <- "sequence"
```

TeDDI

```r
# read feature values for character sequences of the TeDDi sample
teddi.data <- read.csv("~/Github/UpperPalCombinatorics/results/features/teddi_features.csv")
# add column with subcorpus information
teddi.data$subcorpus <- rep("TeDDi", nrow(teddi.data))
# select columns
teddi.selected <- select(teddi.data, c('subcorpus','text', 'num.char','huni.chars', 'hrate.chars',
                           'ttr.chars', 'rm.chars'))
# rename column with sign sequences
```

```
names(teddi.selected)[names(teddi.selected) == "text"] <- "sequence"
```

Combine

Combine the different data frames.

```
estimations.df <- rbind(sign.data.selected, cuneiV.data.selected,
                        cuneiIV.data.selected, cuneiIII.data.selected,
                        teddi.selected)
head(estimations.df)
```

```
##     subcorpus       sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| ////// _ V       12  1.8962406   1.5295563 0.41666667
## 2 Aurignacien          vvvvv        5  0.0000000   0.5711903 0.20000000
## 3 Aurignacien      vvvv vvvv        9  0.5032583   0.9060047 0.22222222
## 4 Aurignacien vvvvvvvvvvvvvv       14  0.0000000   0.6379550 0.07142857
## 5 Aurignacien            |||        3  0.0000000   0.4308271 0.33333333
## 6 Aurignacien           ||||        4  0.0000000   0.5070802 0.25000000
##    rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Simple Stats

```
nrow(estimations.df) # number of sequences with feature estimations
```

```
## [1] 3031
```

```
unique(estimations.df$subcorpus) # names of subcorpora
```

```
## [1] "Aurignacien"          "Cuneiform (Uruk V)"   "Cuneiform (Uruk IV)"
## [4] "Cuneiform (Uruk III)" "TeDDi"
```

```
nrow(estimations.df[estimations.df$subcorpus=="Aurignacien",]) # number of Aurignacien sequences
```

```
## [1] 211
```

```
nrow(estimations.df[estimations.df$subcorpus=="TeDDi",]) # number of TeDDi (modern writing) sequences
```

```
## [1] 995
```

```
nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk V)",])
```

```
## [1] 111
```

```
nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk IV)",])
```

```
## [1] 859
```

```
nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk III)",])
```

```
## [1] 855
```

Write to file

```
write.csv(estimations.df, "~/Github/UpperPalCombinatorics/results/features/features_combined.csv", row.r
```